

Talha Yildirim

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EDUCATION

University of Waterloo

Waterloo, ON

BMath, Computer Science and Applied Mathematics

Expected Graduation: May 2027

- **Relevant Coursework:** Distributed Computing, Networks, Operating Systems, Data Structures, Algorithms, Compilers, Databases, Object Oriented Programming, Cryptography, Concurrency and Parallelism

EXPERIENCE

Software Engineer Intern

May 2026 – Present

OpenText

Waterloo, ON

- Deploying a **Rust** log-parsing binary on **AWS Lambda**, triggered via **S3** events, increasing efficiency of an **inference pipeline** by reducing **LLM** cold start latency from **36 min** to **23s**, with a projected **\$7k/month** in cloud savings

Software Engineer Intern

Jan 2025 – Apr 2025

Sun Life

Toronto, ON

- Implemented **Kubernetes** pod and resource quotas across namespaces using **Helm** and **Kustomize**, optimizing cluster utilization based on real-time metrics, reducing compute costs and overprovisioning by **30%**
- Setup infrastructure and IAM roles for ECR lifecycle policies, using **Terraform**, saving **\$1000/month** in server costs
- Built a containerized **Go** consumer that ingested **Snyk** vulnerability findings from a **Kafka** event stream and automatically tagged Git commit status to enforce security gates, blocking vulnerable merges in **500+ PRs/month**

ML Infrastructure Engineer

Sept 2023 – Feb 2025

WAT.ai

Waterloo, ON

- Improved data pipeline efficiency by **40%** via **GitHub Actions** by automating the preprocessing of raw data, filtering key parameters, handling missing values, and aggregating features into vectorized format, using **Python** and **Pandas**
- Exposed wildfire model as a REST API via **AWS API Gateway** and **Lambda**, ingesting real-time weather telemetry and returning zone-level burn forecasts for **36 regions**, enabling on-demand querying for downstream services

Platform Engineer Intern

Jan 2024 – Apr 2024

Questrade

Toronto, ON

- Orchestrated a series of DAGs in **Apache Airflow** via **GCP Composer**, to validate data integrity between staging and production **BigQuery** datasets nightly, with Slack alerts for anomalies, reducing data quality issues by **60%**
- Deployed a **Kafka** cluster to a production environment through Harness, using **Kubernetes**, **Helm**, and **Makefiles**
- Developed an **RBAC** system using **Django** REST API endpoints and a CLI tool to address growing security and access management needs, resulting in enhanced system security and streamlined user role management

Software Engineer Intern

May 2023 – Aug 2023

Royal Bank of Canada

Toronto, ON

- Spearheaded a project to clone and manage **AWS Terraform** policies by building CI/CD pipelines with **Jenkins**, incorporating automated testing to ensure security compliance and compatibility of modules via testing units
- Wrote a **Bash** that scrapes **SCCM**, dispatching weekly server health digests over SMTP, reducing task time by **90%**

PROJECTS

Parallel-Logram | *Rust, Rayon, DashMap*

- **Rust** implementation of the **Logram** log parsing algorithm, parallelizing **n-gram** dictionary construction via **multi threading** and **concurrent hash maps** to extract static and dynamic tokens from large unstructured logs datasets
- Benchmarked performance with **Hyperfine** across 8+ real-world log datasets (HDFS, Spark, OpenStack), achieving a **4.9× speedup (4.6s to 934ms)** and validating correctness and scalability against the original [academic paper](#)

CUDA CNN Accelerator | *Rust, CUDA-C, RustaCUDA*

- **Rust**-based **CNN** inference engine, with dual **CPU/GPU** backends to quantify **GPU** parallelism gains
- Optimized the build system using **Cargo's build.rs** to compile **CUDA-C** kernels **ahead-of-time** to **PTX**, caching compiled artifacts and triggering recompilation only on kernel changes reducing iteration time by up to **68%**

TECHNICAL SKILLS

Languages: Rust, C, C++, Go, Java, Python, JavaScript, TypeScript, Bash, Terraform

Frameworks: Redis, PostgreSQL, MongoDB, FastAPI, Django, Node.js, Actix, gRPC, Heroku, NGINX, Jenkins

Technologies: Docker, Kubernetes, AWS, GCP, Prometheus, Grafana, Apache Spark, Apache Kafka, Git, Linux, tmux